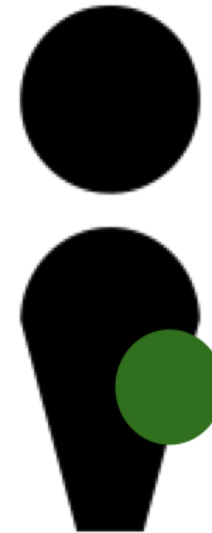


Patient-centered oncological care

New paradigms in understanding and treating cancer



Corresponding Lead: Dr. Irene Grossmann

Planned start date of seed funding: 1-9-2022

Primary Convergence theme: *Technology Supported Transitions in Health(care)*

Secondary Convergence theme (if applicable): *Improving Health Journeys*

Tertiary Convergence theme (if applicable): *Fundamentals of Health & Disease*

Quaternary Convergence theme (if applicable): *Human Centered Technology and AI for Health*

Keywords: *Oncology, safety, game-theory, AI, shared decision-making, personalized medicine, inclusive research*

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1.b Summary

At the heart of treating cancer patients lies an understanding of tumor behavior, as well as an understanding of patients' objectives and their physical tolerance for treatment. The truth is no tumor is or behaves alike. Neither does the average patient exist. Our proposal anticipates this reality by taking the individual tumor and patient as the starting point in treatment design.

We combine several techniques to design the optimal treatment for an individual patient. First, we embrace the fact that a tumor is a dynamic entity that responds to its environment, both host and treatment. This dynamic interaction can be simulated, and thus predicted, using a.o. evolutionary game theory. Omics and (AI derived) histopathological data will be integrated into these simulation models to assess response to systemic and radiation therapy. The generated model will be (iteratively) tested and optimized using in-vitro cell cultures derived from the individual patient's tumor. Advanced risk assessments such as Bayesian networks are applied to assess individual risks in on/off situations such as when surgery may be part of the treatment. Patient objectives and preferences are integrated into a decision-model that supports both patient and clinician.

Succeeding in thus improving the quality of oncological care, will change the standards of medical research in oncology. By demonstrating that the complexity of reality is not a burden but instead a source of information that can be used to the benefit of care delivery, we show that a vastly more inclusive design is attainable and of added value. And it is all done within existing treatment resources. Our innovation lies in the mathematically outsmarting of cancer.

1.c Public Summary

At the heart of treating cancer patients lies a need to understand both the disease and the patient. The truth is no tumor is or behaves alike. Neither does the average patient exist. Our proposal anticipates this reality by taking the individual tumor and patient as the starting point in treatment design.

We combine several techniques to design the optimal treatment for an individual. First we use mathematics to help us predict how the tumor will develop and how it will respond to treatment. It was observed that cancer has typical patterns behaviour that can be mimicked in mathematical models. One of them is 'evolutionary game theory', that mimicks the flourishing or death of organisms (such as a cancer growth) depending on certain characteristics (such as certain cancer-cell types) in specific environments (responses on certain drugs or radiation). To check whether the models can really predict the cancer cells responses, we simulate this interaction in a glass dish before treating the patient. Some tumors can also be treated with surgery, but that comes with risk on functional harm or death. To decide whether it is worth taking this risk, an individual risk is calculated through Bayesian network modelling. And, what an individual wants and needs from a personal point of view, will be taken into account already when the treatment is being designed, not afterwards. This way the treatment will fit the individual the best way possible.

This is a new way of looking at cancer. It is not an easy way, but with the new technologies and new understanding of cancer and patients, we believe it is possible to design better treatments that are less harmful, by being smarter than the cancer.

Relevance to Convergence Health & Technology mission

2.a Contribution to H&T mission and fit with H&T theme(s)

The core of our proposal on personalized oncological care is **integrating** knowledge from different domains to design better and safer cancer treatments within existing treatment resources. This answers fully to the Convergence Health&Technology mission.

Up till now, cancer is very much regarded a static entity that is bombarded with aggressive treatments to eradicate all cancer cells. With that, a lot of healthy tissue is removed or damaged as well, leading to functional disorders and/or long-lasting compromise in health and well-being. It is known that many treatments grossly ‘overdo’ it, but it is not known how treatments could be applied in more strategic sequencing, timing and dosing, to be more effective and less harmful for the patient. This is in part due to the fact that cancer is up till now viewed within one paradigm; as a growth that requires eradication, limited only to what the host can maximally endure. But what if looking from a different perspective could open the door to **outsmarting cancer**?

The core of our proposal does exactly that: it looks at cancer from a different paradigm. How this answers to the Convergence vision can be described as follows:

- It applies, previously unknown to medicine, dynamic mathematical modelling of **cancer behaviour** and tumor-host-treatment interaction via e.g. game-theoretical and Bayesian risk modelling. These models integrate emerging medical knowledge, such as AI-derived predictive histopathological factors and omics data. This is a clear example of tearing down the boundaries between disciplines.
- It embraces the fact that **every patient is different and no cancer growth is alike**. So instead of creating knowledge on the average cancer in the average patient, it takes the individual patient as the starting point. Predictive assessments will be based on the individual tumor and individual patient and, in contrast to standard methodology, will keep assessing the patient/tumor characteristics in the course of the treatment. Treatment designs will be individually tailored to create a better fit and reduce unnecessary harm. The patient is involved in deciding on their treatment objectives and choices right from the start. This follows the aim to create individually tailored patient-centered precise medical treatment for all.
- In a living lab approach^{3,24} (junior) researchers, clinicians and patients will continuously interact to find solutions to actual clinical questions. Students are very welcome and invited to think along and move between the various disciplines involved. This inspirational environment will be an example of how changing thought-perspectives can lead to significant improvements, even within the scope of currently available treatments. Our aim as in the H&T vision, is to inspire researchers to learn from each other’s insights and skills.

Our proposal fits in multiple convergence themes. It leads to understanding cancer and the dynamics of cancer behavior and therefore fits into the theme “ Fundamentals of health & disease”. It fits in “Human-centered technology and AI in healthcare” because we will use AI techniques both in our data-driven predictive models and risk assessments. We will improve health journeys, through active involvement of patients and their objectives and preferences in treatment designs. Last but not least, we will apply advanced technologies and data processing techniques in designing personalized treatments. This project will lead to a true transition towards personalized oncological care, a transition that is both wanted and needed.

2.b Synergy between disciplines

The synergy between disciplines is met on multiple domains and is a strength in our proposal.

One, already mentioned, is applying dynamic mathematical modelling in the understanding of cancer behaviour and cancer-treatment-host interactions. Clinicians are often puzzled by cancer behaviour and are regularly confronted with the feeling that ‘the moment they think they understand, the cancer does not do what they expect’. This signals that we do not grasp how cancer behaves. Viewing cancer as a ‘living entity’ that is in constant interaction with its environment, that responds to treatment and its host, and thus modelling this through evolutionary game-theoretical approaches, has already begun to create a better understanding and more effective treatments ^{16,19,25-27,38,41,42,55-57}. This is an example of how clinical - and mathematical knowledge meet and explore opportunities that arise from this synergy. A further step we take is integrating emerging predictive techniques such as AI derived histopathological features and omics data into the mathematical models ^{8-10,17}. This truly crosses boundaries, because scientists in the field of bio-informatics rarely had knowledge on mathematical modelling and vice versa. But the truth is, that the lack of breakthrough success of advanced predictive modelling thus far, may lie in throwing away information that lies within this integration, within *combining* these data and knowledge. All these steps focus on improving predictions on expected cancer behavior, but we also need to know whether our predictions make sense for the individual patient. This is why for each patient and every tumor, the expected response is tested in an in-vitro model ^{40,58}, that will instantly give feedback on the value of our predictions and serve as input to optimize our modelling. An outcome of the synergy between these disciplines is that we can design personalized treatment and take the concept from theory into clinical practice.

Another synergy lies in incorporating patient-related factors into the treatment design. There is a wide range of valuable information from individual patients that need to be incorporated. One is what patients can endure, in the instances that high-risk surgery is an option in treatment; do the risks of surgery weigh up to the benefits in *this individual* patient? Bayesian risk modelling can contribute to this relevant and mostly unanswered question up till now. But also, what are the patient’s objectives and preferences, what do they seek? How are decisions made and what weighs into those decisions? And how do we translate this information into treatment designs, instead of discussing this afterwards? Here psychology, multi-criteria decision-modelling, communication and shared decision-making and organization of care work in synergy to create a clinical environment that support individualized choices.

And it does not end there yet. Our vision is to create a medical approach to research and standards of care that serve the individual patient. For instance, clinical research ought not exclude specific complex patients from trial populations, but instead **embrace** individual differences to create individualized interventions. This is in keeping with ongoing debates about the changes needed in medical research paradigms, to be more inclusive and to better integrate evidence from observational and pragmatic trials, as well as basic scientific reasoning and knowledge about local disease mechanisms ^{2,53}. The work done in this project will thus contribute to debates in philosophy of medicine, research ethics, guidelines development work on the integration of plural sources of evidence, and clinical trial designs. Developing new research and treatment paradigms also has practical implications, such as adoption issues amongst healthcare workers and patients. To ensure acceptance and implementation of our approach this issue is taken early on and throughout the research program, with close interaction between the clinic and our experts in that field.

2.c Long term sustainability plan

Our proposal integrates a broad range of disciplines to work towards one goal and is very ambitious in that sense. This is the strength of our proposal, at the same time the risks hereof are evident. To ensure our synergistic approach remains feasible, we limit our clinical focus on few specific use cases, even on specific clinical questions within these use cases. From these ‘sample cases’ we hope and expect to formulate general principles that can successively be developed for a wider range of cancers and situations. Our proposal is designed to delve deep into the integrated concept and ‘play’ with all available data to find scientifically and clinically suitable approaches. Our concepts truly are innovative and even if the first steps were already successful^{25,42,55,59}, it carries the risk we will not always succeed and we will likely meet unexpected barriers. We do not see this as failures, because it will mean an increase in knowledge that will only teach us more about cancer and cancer behaviour. In our group there is room for open debate, for failing and reporting unsuccessful attempts, thereby creating and sustaining a learning environment⁷. This first 5-year period is thus meant to explore approaches for precision medicine on highly concrete clinical questions within oncological care, that in the future will be continued for other cancers or even other medical conditions. We aim to maintain our strong core group in a new virtual research and teaching institute (see next section) to disseminate our philosophy and knowledge and seek national and international cooperation. We aim to formalize and structure our collaboration, including erecting a steering group and a governance board.

2.d Visibility, attracting talents and new partners

We intend to erect a multi-institutional centre for mathematical modelling for precision medicine (working title Maths4Health). This will be a platform where members of participating groups meet, discuss and formulate new research projects. The objectives will be to keep the interdisciplinary cooperation going by working together in joint research and education, to keep learning from each other and constantly seek for opportunities how the synergy between domains can answer societal questions. It will use existing multi-institutional structures and in turn facilitate those, for instance through offering electives and involving students through BSc&MSc research projects. We will create an attractive digital platform that explains the research and possibilities for involvement, including for international students. All participants in our group are dedicated to involve and support students and junior researchers in their work, as also shows in their respective teaching efforts (see also section 4.a). This platform will offer opportunities for thesis projects and will be accompanied by teaching for all three (or more) institutions. Several multi-institutional courses will be offered, such as a newly elected Master elective on Health Systems that will be offered at TU Delft led by three teachers/developers from our research group (KS, IG, SC). Also existing multi-institutional, multidisciplinary programs such as the Nanobiology program and the MGC program course on single cell analysis (both MPC) and MSC course Ethics and Healthcare Technologies (SC) and the BSc program Health Sciences at the Erasmus University (MdM) will be advocated on our virtual platform. Students from all disciplines (technical, medical, social) will be welcomed and stimulated to work together across disciplines on projects within our centre. Regularly scheduled meetings, roundtable discussions and seminars will be organized. We constantly seek for partners that have an active involvement, as we have successfully done so far in preparing our proposal. Funding will be sought in national and international funds (NWO, Horizon, ZonMw) and industry partners. The platform will be a means to stay in touch, with each other and the world around us.

2.e Long term societal impact

The most pronounced contribution to healthcare will be the design of better and safer treatments in cancer. Cancer is the second leading cause of death (WHO) and a significant cause of human suffering around the globe. We aim to ease that suffering by removing ineffective and harmful static treatment regimens from the clinic and replacing them by more tactical therapies tuned to individual patients' cancer and needs.

It is expected to have a positive economic impact, because although the staging phase is more elaborate, the treatment will be using up less resources (by using these more smartly) and the treatment will be less toxic, using up less toxicity treatments. Patient safety is an active component in our approach by establishing more realistic risk estimates that help doctors and patients decide to actually do less. Because even when doctors are aware some risks are high, as long there is a perceived high uncertainty (and thus a chance it may well go right) and no felt real alternative, patients usually accept high risks. If more realistic estimates on risk combined with better predictions on expected outcomes from other treatment-scenarios can be shown and balanced, better choices can and will be made.

Calls for revising contemporary models of evidence-based medicine are increasing, as the need to accommodate individual patient complexity and to utilize available data about individual patients are both urgent and key to improving the effectiveness of medical treatments, both in terms of health outcomes and costs². What is called for is a more person-centered, bottom-up approach such as will be demonstrated in our project. Beginning with the individual patient offers an opportunity to develop new ways of incorporating not only diverse biological profiles of individual patient, but the diverse needs, capabilities and values of different patients in the very early stages of treatment design. Our approach is a concrete example that answers this call.

The disciplines that will benefit from this research are (mathematical) oncology, cancer biology, and medicine in general. We set a new paradigm in treatment design, by taking our bottom-up personalized approach into the clinic. We take all that matters to ensure a successful implementation into the proposal, including organization & infrastructure, adoption issues, patient involvement & co-creation and of course in-depth understanding of what the actual clinical problems are that need to be solved. We thus integrate expertise and knowledge from very different domains to achieve one thing, and that is getting patients a better and safer treatment. We hope that this inspires other medical domains and offer experience and knowledge from our pioneering efforts to achieve our goals.

The organizations profiting from this research will be universities, hospitals, but also patient advocacy groups, whom we will keep actively involved for the maximum benefit of the patient. Patient organization Inspire2Live will be our active collaborator during this project and beyond.

Standard dissemination strategy will be publishing our results in peer-reviewed journals and publishing all codes/anonymized data created a dedicated website and Github. Our team will also write popular pieces and talk to media about the research and results of our work.

Within the course of this project, we will also not only invite patient advocates to our steering/advisory meetings, but will also actively participate in conferences and events organized by the patient advocacy group Inspire2Live and stimulate their active involvement in our research.

Besides that, the applicants actively collaborate with the Moffitt Cancer Center who the world leader in the evolutionary therapy. Interactions between the team and collaborators from Moffitt and other members of the Oncology Research Information Exchange Networ (ORIEN) will lead to better results and improved patient-centered care in cancer worldwide.

Within this project, we will set up clinical trials on evolutionary therapies in different types of cancers, based on the evidence collected during this project. We will also expand the patient dataset for that of other important organizations, such as the EORTC Lung Cancer Group, chaired since the past year by Anne-Marie Dingemans.

All our results/codes will be published open-source so that any researchers, patients, and practitioners can benefit from these results and can contact us for possible future collaborations/advise.

Flagship proposal

3.a Ambitions

Our ambition is to build an integrated, comprehensive method for personalised oncological care.

The innovation lies first and foremost in understanding and predicting cancer behavior, using game-theoretical mathematical modelling. Envisioning the *dynamic* cancer behavior, such as in growth - and metastatic patterns and in how it will respond to systemic or radiotherapeutic treatment. It also lies in embracing the reality that no tumor or patient is alike, and *using* this knowledge in personalised risk assessments and designing individualised care.

The challenge in our proposal is the integration of knowledge and methods from different domains, that in our vision can together create an understanding of cancer where single-substrate strategies cannot. Then, to succeed in implementing such personalised oncological care, more than just technical skills are needed. It requires adoption by care professionals, creating care pathways that support individualised treatments, and a (more) central role for the art of shared decision-making. Such an integrated approach, as is summarised in Figure 1, may be key to advance personalised medicine from the experimental to the clinical domain.

This ambition is fully in line with the ambitions from Convergence, on aspects of integrating domain-specific knowledge and connecting researchers from very different backgrounds into one goal: making sure that the individual cancer patient receives the best and safest treatment possible.

Clinical care pathway

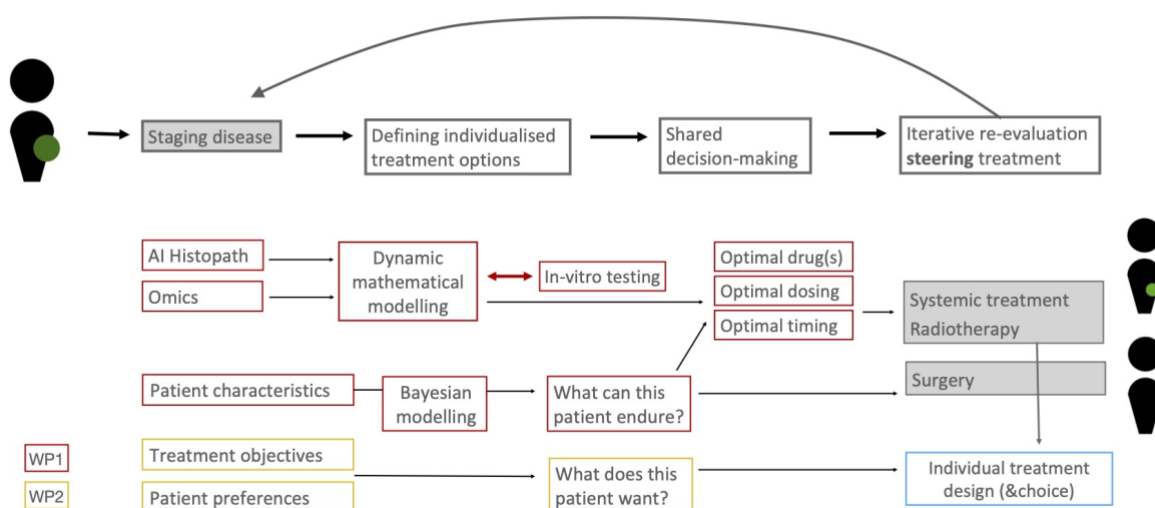


Figure 1: The envisaged clinical care pathway

3.b Aims and feasibility

Program outline: The aim is to develop novel mathematical models that can predict tumor behavior, both anticipating on disease development and treatment response. These models are built from existing retrospective patient and tumor databases, and validated through prospective in-vitro experiments and observational studies to test the model robustness. Because it will result in a personal treatment design, patients may be faced with multiple options and also likely treatment-changes over time due to dynamic tumor behaviour. The character of consultations will thus change, patient engagement will be even more important as will shared decision-making. To anticipate on that, preparations will be made to create a fitting clinical care environment. Parts of the envisioned pathways can be simulated as pilot studies in current oncological care. When the models are built and the new care pathways embedded, clinical trials will start. Impact evaluation will be a continuous process throughout the program, focusing on a.o. the need and difficulties of developing personalized (inclusive) approaches. Adoption of these new paradigms needs appropriate attention to ensure the success of implementation. In all of these work packages students are welcome and will be invited to work with us; see also section 2.d.

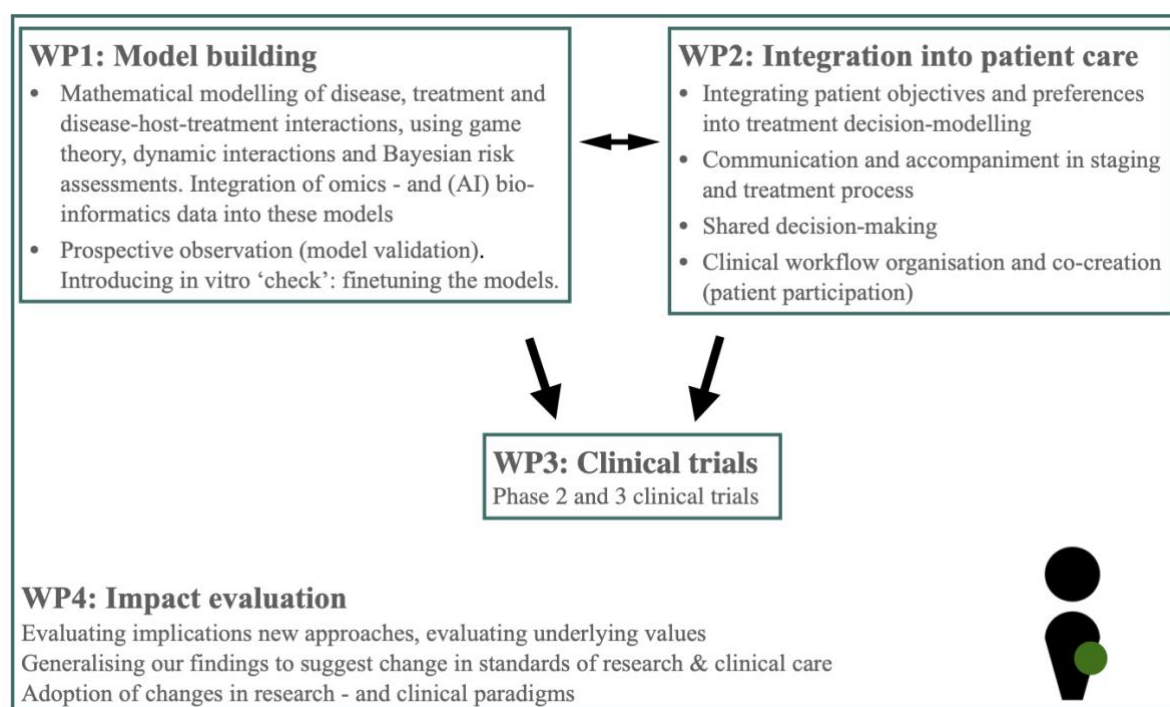


Figure 2 Depiction of the program outline (see also section 3.c)

Feasibility: Based upon previous work concerning modelling dynamic disease behaviour, it is estimated that the clinical application will be feasible after two years. The models will iteratively be improved in clinical practice, that will cover year 3 till 5. To keep the program feasible, we choose to work on four distinct use cases (see section 3c). The use-cases have similar work-up plans and the researchers will work on all of them, also to increase our knowledge on differences and similarities between tumor types. We expect the model-building concepts will be generalizable to other tumor types. For financial feasibility we refer to the budget plan.

Risks

- The main uncertainty in our study lies in the time needed to develop a valid and robust model to implement in a clinical trial. The *impact* of this risk is limited; the work done in WP1 is already very valuable in creating a better understanding of cancer behaviour in relation to its histopathological and -omics characteristics. The risk can be mitigated by ensuring a longer term collaborative, as we will establish.
- A practical obstacle may be the problem of misrepresentation of the tumor heterogeneity through sampling techniques, such as is already anticipated upon in the use case rectal cancer. This can be solved through performing an additional intermittent clinical study for biopsies optimization (which is feasible to do).
- Because the application of Bayesian networks for risk assessments has not yet been thoroughly studied in the medical domain, it is unknown so far whether the predictions are precise enough for use in clinical practice. If we encounter this problem, we will carry out studies to define the data needed and set in motion that these data will prospectively be collected on a national level, using existing infrastructure.
- Patient compliance may be compromised by the dynamic treatment protocols, in the sense that it may be experienced as more uncertain due to the iterative adaptations during treatment. This can be resolved by identifying and addressing these concerns in the consultation room or in other words, incorporating them in the patient's preferences and objectives. The involvement of patient-advocates in our research will help to anticipate on that.
- As a result of the prolonged staging of the disease, a delay in time from diagnosis till treatment may occur. The impact of possible delay will have to be weighed against the benefit of an individualized treatment scheme.
- Implementation may be hampered due to the highly innovative and complex character of our approach. This is the reason why 'adoption' is an important topic in our proposal.

Future external funding

is already set in motion and includes applications for:

- NWA-ORC Route Personalised Medicine 085
Initiative text is submitted. Supported by the TU Delft innovation&impact centre.
- NWO VIDI (KS, MPC this year)
- NWO Open Technology program (MPC)
- KWF-TKI (MPC)

And intention to apply for grants such as mentioned in section 3.d

3.c Work plan

Underneath the work plan from each workpackage is described. All research operates within existing infrastructures and do not need additional facilities.

WP1: Model building: Recently, members of the WP1 team designed the first mathematical models guiding the so-called evolutionary therapy, i.e. therapy that anticipates and steers cancer cells' response to treatment, in clinical trials^{4,42,55,56,59}. Building further upon this approach, following use cases were selected for further development:

- metastatic non-small cell lung Cancer (NSCLC) with driver mutations, addressing the specific problem of evolution of therapy resistance^{6,13,43}
- neuro-endocrine lung cancers for the property of difficult determination of histopathology and therewith treatment choices¹⁵
- oro/hypopharyngeal/lanryngeal squamous cell carcinoma because of the lack of knowledge on which treatment approach is the best, also in relation to compromises in oropharyngeal function and subsequent decrease in quality of life^{5,12}
- rectal cancer for the property of high surgical risks and the emerging alternative of organ-saving therapy that requires better estimates on both surgical risk and expected treatment response^{20,35}

We will extend classical ecological models of tumor growth¹⁹, by including evolutionary dynamics of cancer cells in response to treatment. We will utilize large datasets of retrospective in-vivo data for initial model fitting. These initial models will be tested and validated through in-vitro functional assays (including cell susceptibility and resistance to treatment and their correlated (single cell) omics data – DNA or RNA)⁴⁰. Aside of standard volumetric information, we will include omics and histopathologic data as additional inputs, through machine learning (ML) techniques: (i) Convolutional Neural Network methods for extraction of histopathological features and their integration with multi-omics features, (ii) supervised ML methods for identifying cancer cell types with known profiles, and (iii) Deep Neural Network methods for learning unknown profiles. Our preliminary research demonstrates how to combine these methods into GT models¹⁷. Each patient/tumor will have their own parametrization of the model. When the models are validated through both in-vitro and in-vivo data, we will utilize multi-objective optimization considering different treatment targets, e.g. time to progression, treatment toxicity, or total tumor burden, to find timing/dosing of available therapies. A Bayesian risk assessment³⁷ will be conducted for the *patient* using existing data, to find the relationship between their individual characteristics and the risk of treatment quantified by morbidity and mortality or survival.

The outputs of WP1 will be *personalised treatment schemes*, outlining, where applicable, multiple 'best treatments' that will allow the patient and physician a tradeoff among the relevant criteria.

WP2: Integration into patient care: To support this personalized approach, care-pathways will be redesigned during co-creation workshops involving all stakeholders. We will use the new method Metro Mapping^{21,22,36}, developed to support shared decision-making in oncology^{11,44-45}. It involves redesigning care-pathways, and optimizing information provision to the patient, team roles and responsibilities. Unlike traditional care-protocols that are meant to uniform care, our protocols are designed to support personalized care. In order for the model to optimally fit work practices, a selection of existing care-pathways will be studied and redesigned through document analysis, observational studies, complemented with interviews or focus groups. For both patients and professionals, educational materials will be developed. The models and pathways will be studied in clinical situations that approximate personalised care best. Such a personalized care trajectory,

including the process of shared decision making, will then be evaluated from both patients' and professionals' perspectives.

WP3. Clinical trials: This work package will consist of the clinical studies guided by our clinical leads. The models and work methods will continuously be improved. Inherent to the personalised approach, a randomized trial will not be possible. Instead, we will utilize observational prospective matched-control studies to demonstrate safety and efficacy of the novel treatments. In close interaction with methodology experts from WP4 an appropriate scientific trial design will be defined.

WP4: Impact: Our approach was born out of a vision to create inclusive medicine, by looking at research methods and treatment designs that respect individual differences. WP4 will consider, adopt and potentially design appropriate clinical trial formats that produce robust and generalizable evidence meeting the demands of personalized approach. The additional aim of this overarching work package is to look closely at the paradigm shift of medical professionals, which is essential for *implementation* of our approach in clinical practice. We will map facilitators and barriers for adoption of our approach, such as those related to professional norms and values, the patient-doctor relationship, dealing with uncertainty, and trust in new technologies. Since the wider context of innovation also serves as a facilitator or barrier, we will take organisational, financial, ethical and legal aspects into account.

3.d Alignment of the program with relevant national and international research agendas

Our proposal is on many aspects very much in line with research national and international research agenda's: On cancer in general, on personalized medicine, on applying artificial intelligence techniques into clinical applications, on predictive biological markers, on patient involvement, engagement and shared decision-making amongst. Underneath several examples are listed.

Relevant NWA-ORC research agenda questions

085: Every tumor is different: how can we understand cancer well enough to develop a treatment for each form?

095: How can healthcare, for example by using biomarkers, be more focused on the uniqueness of a person?

081: How will knowledge of genetics play a role in understanding, screening for and treating (rare) diseases?

104: How do we develop minimally invasive techniques and interventions for the diagnosis, prognosis and treatment of patients?

Relevant EU research agenda examples

Horizon Mission on Cancer: Research and Innovation actions supporting the implementation of the Mission on Cancer

Horizon Europe Global Challenges: Computational models for new patient stratification strategies

Horizon Health: Clinical validation of artificial intelligence (AI) solutions for treatment and care

Dutch Cancer Society - OncoCode Institute

Outsmart cancer

Zorginstituut Nederland

Citizen- and patient participation

Shared Decision-making

Research anticipating on personalised medicine

Ministry of Health

Knowledge agenda Right Care in the right place

ZonMw

Program on outcome-oriented care, with special attention for the prerequisites for shared decision making in clinical practice.

Not covered by research agenda's (yet)

International networks have recently begun to tackle the problem of evidence in medicine from philosophical and additional perspectives (e.g., in Europe: CauseHealth, EBM+, CEBM, PhilPharm), but there have been few opportunities to consider the epistemology of evidence integration within innovative medical practice, a gap this project can fill.

Consortium

4.a Track record of the consortium members

Dr. Katerina Stankova (KS)^{41,55} TUD associate professor and Delft Technology Fellow at Delft University of Technology. She leads a number of national and international projects in game theory and mathematical oncology, such as EU ETN and NWO KLEIN2 projects. KS won the 2018 Integrated Mathematical Oncology prize and 2020 NWO Stairway to Impact award for designing novel cancer therapies and bringing them to clinical trials. KS has been extending and validating data-driven mathematical oncology models, to improve their predictive capabilities^{4,19} and to guide evolutionary therapies for more aggressive cancers or cancers with not many treatment options, such as those treated within this flagship³⁸. She works closely together with co-applicants **Dr. Monica Salvioli** and **Dr. Johan Dubbeldam**.

Dr. Miao-Ping Chien (MPC)^{40,58} EMC assistant professor, PI at Oncode Institute, head of UFO Biosciences (microscopy-based functional single cell screening and sequencing). Multidisciplinary scientist in molecular biology, (bio)chemistry, advanced imaging, computational biology and single cell biology. Invited as a keynote or plenary speaker in renowned conferences (e.g. VIB, Oxford Global). She has two filed patents and has been honored with international awards and received multiple grants (e.g. NWO Veni, Cancer GenomiCs.nl (CGC) Junior Fellowship, Oncode Institute Junior Fellowship, KWF, Ammodo Science Award, Erasmus MC fellowship, Technology Development Fund) since she started her own research group at Erasmus MC in June 2017, totaling over 3 millions of euros.

Dr. Yunlei Li (YL)²⁹ EMC assistant professor on the design and application of novel machine and deep learning methods to integrate multiple data modalities including omics, clinical and imaging data for clinical cancer research and clinical trials. Funded by national (KiKa, KWF, NWO, ZonMW, Hanarth, Stichting Lever Onderzoek) and international sources (EU Horizon, EU FP7, EU Eurostars), she is contributing to battle leukemia, pancreatic cancer, lung cancer, and more diseases. Dr. Li supervises 4 PhDs, 1 postdoc and 6 MSc/BSc students.

Dr. Jan von der Thüsen (JvdT)⁴³ EMC associate professor and deputy head of department in pathology, lead thoracic pathologist and molecular diagnostics liaison at Erasmus MC. Research interests include the pathology of thoracic diseases and AI in histopathology (supervising 12 PhD students). He is in multiple steering boards such as; pathology chair of the EORTC lung cancer group, medical expert for the ESP lung EQA scheme, pathology committee member of the IASLC, and board member of the Dutch Lung Cancer Audit (DLCA).

Dr. Amir Pooyan Afghari (APA)^{1,37} TUD assistant professor on safety science and probabilistic risk modelling. In 2016 he received an Outstanding Paper Award in the United States. He has supervised several graduate and post-graduate students and contributes significantly to safety science education. He works with IG on translating safety science methods, such as probabilistic modelling, for healthcare purposes, in close cooperation with **Prof. Dr. Geurt Jongbloed** and **Prof. Dr. Pieter van Gelder**; TUD full professor and section head safety & security science. Research focuses; safety of systems; multivariate optimization theory; Bayesian modelling. He published nearly 300 articles and contributed to more than 10 books. He teaches and supervises multiple BSc, MSc and PhD students from all over the world.

Dr. Jafar Rezaei (JR) TUD associate professor on decision-making methods, including the best-worst method (BWM) that has become mainstream³⁹. Recognized as an influential scientist in this

field, published over 100 articles, functions as editor-in-chief and in multiple editorial teams, received several awards from leading journals. Teaching; courses on Multi-criteria Decision Analysis; supervision over 100 BSc&MSc theses. Valorisation; organization of several international events including the EURO Working Group on multi-criteria decision aiding in Delft in 2018; several keynotes at international conferences. He supervises nine PhD candidates, some based on NWO and EU projects.

Prof. Dr. Anne Stiggelbout (AS)^{19,21-22,44-45} EUR professor of medical decision-making and scientific research program lead on value-based healthcare of the LUMC. She has received many grants from among others the Dutch Cancer Society and the Netherlands Organisations for Scientific Research (NWO, incl. a VIDI grant) and Medical Research (ZonMw), and recently a Horizon Europe Grant (co-coordinator). She is active in scientific advisory committees and boards and was President of the international Society for Medical Decision Making (2011-2012).

Dr. Marleen de Mul (Mdm)^{47,48,52} EUR assistant professor in eHealth. She is leading a research group on digital innovations in healthcare. Her personal interest is the adoption and embedding of new technologies in health care practice (specifically in data-driven, value-based care pathways) and patient participation in technology development (co-creation). She (co-)authored over 40 publications, and supervised over 120 BSc&MSc theses and 3 PhDs.

Prof. Dr. Ir. Kees Ahaus (KA)^{49,50} EUR full professor and Department manager of Health Services Management & Organization. His research focus is on adopting new technology or innovations by providers/ care professionals and patients/ citizens and on value-based healthcare. He attracted multiple funds from ZonMw and received two awards in recent years. He supervised >50 BSc&MSc students and (co)authored over 100 publications.

Pieter Vandekerckhove (PV)^{47,48} EUR senior PhD student. His research and education focus on the development of a co-design methodology for digital health. Pieter has recently published a book based on his experience with cancer at a young age “The Self Hologram: Awakening a philosophy of life”. He has a special interest in improving the patient experience of cancer experience and builds upon his own experience to do so.

Prof. Jan van Busschbach (JvB)^{11,14,28,30,46} EMC full professor and sectionhead Medical Psychology. Research on; patients reported outcome towards clinicians (also 'value based healthcare'); cost effectiveness analysis and health policy. He teaches in international post graduate health sciences course (NIHES) on measuring quality of life, several health economic seminars for health professionals and supervised multiple MSc theses and has been part of more than 150 doctorate committees.

Dr. Leonieke Kranenburg (LK)¹¹ EMC associate professor and healthcare psychologist. Her research topics include patient treatment preferences and decision making. She is co-project leader of the NOSANO and SANO-2 studies on patient preferences regarding an innovative and-organ sparing treatment of esophageal cancer.

Prof. Dr. Anne-Marie Dingemans (AMD)^{6,15,43} EMC professor thoracic oncology. Research interests in neuroendocrine lung tumors, driver mutations and oligometastatic disease. Chair of the EORTC lung cancer group. She is the lead-author of the revised ESMO guideline on SCLC. She is associate editor of the Journal of Thoracic Oncology and editorial board member of the International IASLC Lung Cancer News. She has (co-)authored more than 230 publications and book chapters.

Prof. dr. C. Verhoef (KV) EMC full professor and department head of Surgical Oncology. Board member of several (inter)national guidelines committees and (international) Oncology & Research associations. Funded by grants (~6M Euro), his clinical and (translational) research activities are in patients with advanced colorectal cancer, melanoma, sarcoma and breast cancer patients. Common focus: Determination of new biomarkers/prognostic factors and new treatment modalities. He supervised over 50 PhDs, published +/- 500 articles and contributed to several books. He works together with **Dr. Michail Doukas**, co-applicant.

Dr Jose Hardillo (JH)^{5,12,58} EMC assistant professor ENT oncology. His research interests and cooperations include a tumor heterogeneity study together with MCP, further he has a focus on organoid studies and immunostudies on nasopharyngeal carcinoma's in multi-institutional collaboration. His group build an information and biobank from 2006 on (RONCDOC) that can be used for model-building on the ENT use-case. Further his group is experienced in value based care studies. He supervises 4PhD and 5MSc students and works together with **Dr. Esther van Meerten**.

Dr. Samantha Copeland (SC)^{2,31} TUD assistant professor on Ethics and Philosophy. A philosopher, she has published on the nature of clinical reasoning and works in research ethics in the context of emerging medical technology development and implementation. Teaching: MSC course Ethics and Healthcare Technologies; supervision students in technology innovation and management; co-PI of the Impulse project, Converging Ethics for Converging Technologies.

Dr. Claudia Werker (CW)⁵³ TUD associate professor. She focuses on inclusive research and innovation in Science, Technology, Engineering and Mathematics (STEM) disciplines. This fits with her overall research interest on how responsible research and innovation (RRI) systems evolve in the digital age. Received various grant from the EU, NWO, and the Deutsche Forschungsgemeinschaft (DFG).

Dr. Irene Grossmann (IG)^{23,51} TUD assistant professor safety in healthcare and oncological surgeon. Her research interest lies in identifying and where possible solving scientific, clinical and organizational causes of harm to patients. She (co)authored over 20 articles, wrote 1 book chapter and (co)-supervised multiple MSc and 3PhD students.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

WP1: To successfully design the novel data-driven mathematical models for predicting therapy response in cancer cells and to optimize treatment and treatment decisions, we need knowledge of cancer biology, expertise in data science, mathematical modelling (including game theory, dynamic systems theory and Bayesian risk modelling), AI in predictive bio-informatics and people who can connect these different domains. All of this expertise, some with leading experts in their fields, is present in the WP1 team, with many of them already working together for a longer time.

WP2: Designing appropriate care pathways is a shared decision-making problem, which requires expertise from oncological aspects of the problem as well as its behavioral, decision-making and organizational aspects. WP2 brings the leading experts in these fields and all stakeholders (including patients) together to realize the aim of this WP, in similar fashion as WP2.

WP3: The clinical leads will work with members from the other work packages on their specific use cases. The clinicians' expertise is on formulating relevant clinical questions, access to patient

data that is needed to build the models and access to the clinical research infrastructure and lead positions in the clinical trials.

WP4: Changing paradigms in research and treatment does not go by itself. Impact evaluation, reflections on research methodologies and adoption issues all come together in this work package, that is chaired by leading experts in the fields of epistemology, inclusive research methodologies and adoption. Because this is an overarching workpackage, all participants will be involved in its activities through science discussion meetings and on-site exchanges.

4.c Involvement of partners

Dr. Rachel Cavill⁸⁻¹⁰, assistant professor at Maastricht University and top expert on AI and machine learning in medicine. She has been working with KS on translating omics data to inputs of mathematical models¹⁷.

Prof. Dr. Dirk De Ruyscher¹³ is a professor of respiratory oncology/radiotherapeutic oncology at Maastricht University, head Clinical Studies of the MAASTRO Proton Therapy, focusing on individualized treatment protocols aimed at tumor containment with minimal toxicity. He has led a number of national and international grants in academia and industry and led more than 40 trials. He is the coordinator of the biobank at Maastricht Clinic with data of more than 10.000 patients, and chairs the Dutch Platform for Radiotherapy and Lung Cancer. Connection with both KS and AMD since 2019.

Dr. Robert A. Gatenby, MD, head of Center of Excellence for Evolutionary Therapy and Department of Integrated Mathematical Oncology at the Moffitt Cancer Center and “father of evolutionary therapy”, with background in radiology who leads a first clinical trials on evolutionary therapy at Moffitt. Connection with KS since 2015^{16,42}

Prof. Joel S. Brown³⁸ is an evolutionary biologist, senior member of the Moffitt Cancer Center, and senior member of its Center of Excellence for Evolutionary Therapy, and a creator of the mathematical modelling of Darwin’s principles of natural selection, the so-called Darwinian dynamics. Connection with KS since 2015^{16,42,53,55}

Tiello Jongmans and **Peter Kapitein** from Inspire2Live will join our team as patient advocates. They will contribute to the research designs from a patient perspective and help us to disseminate knowledge to the public.

Dr. Rani Lill Anjum², Norwegian University of Life Science, Director of CAPS-NMBU (Centre for Applied Philosophy of Science) and CauseHealth (Causality, Complexity and Evidence in health Sciences) – a transdisciplinary philosophy of medicine network.

Dr. Elena Rocca, Associate Professor of Drugs and Safety, Oslo Metropolitan University. Connection with SC since 2015.

4.d Governance, organization and collaboration strategy of the consortium

See section 2.d where we described our strategy for organization and collaboration. A board of governors will additionally be asked to be our critical friends to help us keep our focus and coherence.

I Budget table (including the external funding table)

R&D Overview: kostensoorten per type onderzoek (hele project periode)

[illegible]

II Letters of intent and letters of commitment



Department of Molecular Genetics

*Dr. Molewaterplein 40
3015 GD Rotterdam, the Netherlands
Tel: +31-10-7043073
m.p.chien@erasmusmc.nl*

Addressed to:

Dr. Irene Grossmann

Delft University of Technology, TPM faculty

Email: I.Grossmann@tudelft.nl

2022/03/11

*Letter of Intent
for the
Patient-centered oncological care.
New paradigms in understanding and treating cancer.
PROJECT*

Dear Dr. Irene Grossmann and Prof. Pieter van Gelder,

I, Miao-Ping Chien, in my capacity of assistant professor at Erasmus University Medical Center hereby confirm that Erasmus University Medical Center intends to contribute to the *Patient-centered oncological care. New paradigms in understanding and treating cancer* project, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

Erasmus University Medical Center intends to provide an in-kind contribution of 2 PhDs (one 1 FTE, one 0.2 FTE), 1 postdoc (one 0.2 FTE) and 1 technician (0.2 FTE), representing a monetary value of € 172,811 and further detailed in the project proposal and budget form.



Cancer Institute

LETTER OF INTENT

for the

Patient-centered oncological care.

New paradigms in understanding and treating cancer.

PROJECT

Direct dial 016-70341902

Internal postal address RG-2

Email c.verhoef@erasmusmc.nl

Our reference CVIEB

Date March 14 2022

Re: Letter of Intent

Postal address

P.O. box 2040
3000 CA Rotterdam, NL

Visiting address

Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route

www.erasmusmc.nl

Dear Dr. Irene Grossmann and Prof. Pieter van Gelder,

I, Kees Verhoef, in my capacity of Head of Department Surgical Oncology at Erasmus MC, Dr. Molewaterplein 40, 3015 GD Rotterdam, Netherlands, hereby confirm that Erasmus MC intends to contribute to the Patient-centered oncological care. New paradigms in understanding and treating cancer project, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

Erasmus MC intends to provide an in-kind contribution of the PhD-candidate, representing a monetary value of € 67.000 and further detailed in the project proposal and budget form.

Yours sincerely,


Name: Kees Verhoef
Position: Head of Department Surgical Oncology
Date: March 14 2022

Prof. Dr. C. Verhoef
Prof. Dr. P. Tanis
Prof. Dr. B.P.L. Wijnhoven
Dr. C.T.I. de Bie
Dr. A.R.M. Brandt-Kerkhof
Dr. G.J.H. Franssen
Dr. T.M. van Ginhoven
Dr. L.B. Koppert
Dr. S.M. Lagarde
Dr. E.V.E. Madsen
Dr. J. Rothbarth
Dr. P.C. van der Sluis
Dr. Th. van Dalen

www.erasmusmc.nl

Erasmus School of
Health Policy
& Management

LETTER OF INTENT
for the
*Patient-centered oncological care:
New paradigms in understanding and treating cancer.*
PROJECT

Dear Dr. Irene Grossmann and Prof. Pieter van Gelder,

I, Werner Brouwer, in my capacity of Dean a.i. at Erasmus School of Health Policy & Management (ESHPM) hereby confirm that ESHPM intends to contribute to the *Patient-centered oncological care: New paradigms in understanding and treating cancer* project, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

ESHPM intends to contribute from awarded projects € 186,731 (125,000 Horizon Europe; 32,369 Citrien; 29,362 ZonMw) towards the project costs in accordance with the budget in the project proposal and budget form.

Yours sincerely,



Name: Prof. Dr. W.B.F. Brouwer

Position: Dean a.i.

Date: 13-03-2022

Visiting address
ESHPM,
Bayle, 7th floor
Burgemeester Oudlaan 50
Rotterdam

Postal address
P.O. Box 1738
3000 DR Rotterdam
The Netherlands

T +31 10 408 8525
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Erasmus University Rotterdam





LEIDEN UNIVERSITY MEDICAL CENTER

Department Biomedical Data Sciences

tel. 31-71-5264574
e-mail a.m.stiggelbout@lumc.nl
date March 10, 2022
Subject LETTER OF INTENT

Patient-centered oncological care.
New paradigms in understanding and treating cancer.
PROJECT

Dear Dr. Irene Grossmann and Prof. Pieter van Gelder,

I, prof. dr. Anne M. Stiggelbout, in my capacity of prof. of Medical Decision Making at Academisch Ziekenhuis Leiden/Leiden University Medical Center hereby confirm that the Department of Biomedical Data Sciences of the LUMC intends to contribute to the project "Patient-centered oncological care. New paradigms in understanding and treating cancer project", as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

The Department intends to contribute € 250K in cash towards the project costs in accordance with the budget in the project proposal and budget form (Dutch Cancer Society Grant 12921: "Improving personalised treatment in oncology. Towards shared decisions based on integrated information regarding treatment options, health status and patient preferences.").

Yours sincerely,

Prof. dr. Anne M. Stiggelbout, PhD
Professor of Medical Decision Making
March 13th, 2022

Leiden University Medical Center

Date March 14, 2022
Contact Dr. Kateřina Staňková
Phone +31 (0)646 066 671
E-mail k.stankova@tudelft.nl
Subject Letter of Intent



Dr. Grossmann and prof. van Gelder
Centre for Safety in Healthcare
Delft University of Technology

**Faculty of Technology,
Policy and Management**
Jaffalaan 5
2628 BX Delft

Dear Dr. Grossmann and prof. van Gelder

I, **Dr. Kateřina Staňková**, in my capacity of a member of the Department of Engineering Systems and Services (ESS), Delft University of Technology, I hereby confirm that I intend to contribute to the flagship “**Patient-centered oncological care: New paradigms in understanding and treating cancer**”, as applied for by the main applicant, **Dr. Irene Grossmann**.

I intend to contribute both in cash and in kind, in accordance with the budget in the project proposal and budget form.

The contribution in cash is intended from two projects I applied for: An NWO XS project and NWO VIDI project. The contribution in kind (0.5 fte PhD and 0.5 fte Post-Doc) is from NWO KLEIN2 project that I lead. All these projects focus on improving cancer treatment through data-driven mathematical modelling and are therefore well aligned with the flagship. The PostDoc matched in kind is supervised by me. The PhD student matched in kind is placed at Delft Institute of Applied Mathematics and is supervised by Dr. Johan Dubbeldam and me. I confirm that Dr. Dubbeldam agrees with the 0.5 fte matching in kind of this PhD student.

Your sincerely,

A handwritten signature in black ink, appearing to read 'Kateřina Staňková'.

Dr. Kateřina Staňková
Associate Professor in Dynamic Game Theory
Department of Engineering Systems and Services
Faculty of Technology, Policy and Management
Delft University of Technology



Prof. dr. J.G.J.V. Aerts, Dr. Molewaterplein 40, 3015 GD ROTTERDAM

LETTER OF INTENT
for the
Patient-centered oncological care.
New paradigms in understanding and treating cancer.
PROJECT

Dear Dr. Irene Grossmann and Prof. Pieter van Gelder,

I, prof Joachim Aerts, in my capacity of head of department of respiratory medicine at Erasmus MC hereby confirm that the department of Respiratory Medicine of Erasmus MC intends to contribute to the *Patient-centered oncological care. New paradigms in understanding and treating cancer* project, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

The department of Respiratory Medicine intends to contribute € 175,684 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

The department of Respiratory Medicine intends to provide an in-kind contribution of 0.3 FTE PhD student and 0.5 FTE post-doc, representing a monetary value of € 151,335 and further detailed in the project proposal and budget form.

Yours sincerely,

Name: prof Joachim Aerts

Position: head department respiratory Medicine

Date: 14-03-2022

Letter of Intent for a project partner

www.erasmusmc.nl



LETTER OF INTENT
for the
Patient-centered oncological care.
New paradigms in understanding and treating cancer.
PROJECT

Dear Dr. Irene Grossmann and Prof. Pieter van Gelder,

I, dr. Jose Angelito Hardillo in my capacity as principal investigator and consultant staff head and neck surgeon at the department of Otorhinolaryngology –Head and Neck Surgery hereby confirm that the department of Otorhinolaryngology-Head and Neck Surgery intends to contribute to the *Patient-centered oncological care. New paradigms in understanding and treating cancer* project, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

The department of Otorhinolaryngology- Head and neck Surgery intends to provide an in-kind contribution of 0.20 fte manpower, PhD student/researcher representing a monetary value of € [45.000] and further detailed in the project proposal and budget form.

Yours sincerely,

Name: dr. Jose Angelito U. Hardillo

Position: principal investigators/ consultant head and neck surgeon

Date: 14-03-2022

Letter of Intent for a project partner

www.erasmusmc.nl

Prof. Dr. Dirk De Ruyscher: external partner (matching in kind)



March 14, 2022

LETTER OF INTENT
for the
Patient-centered oncological care.
New paradigms in understanding and treating cancer.
PROJECT

Dear Dr. Irene Grossmann and Prof. Pieter van Geider,

I, Prof. Dr. Maria Jacobs, in my capacity of CEO at Stichting Maastricht Radiation Oncology (Maastricht), hereby confirm that Maastricht intends to contribute to the *Patient-centered oncological care. New paradigms in understanding and treating cancer* project, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

Maastricht intends to provide an in-kind contribution of 0.2 FTE of Prof. Dr. Dirk De Ruyscher, representing a monetary value of 162 k€ per year and further detailed in the project proposal and budget form.

Yours sincerely,

A handwritten signature in blue ink, appearing to be 'Maria Jacobs', written over a horizontal line.

Name: Prof. Dr. Maria Jacobs

Position: CEO Stichting Maastricht Radiation Oncology

Date: March 14, 2022

Dr. Tanslaan 12
6229 ET MAASTRICHT

+31 (0)88 44 55 600

info@maastro.nl
www.maastro.nl

Postbus 3035
6202 NA MAASTRICHT

Dr. Rachel Cavill: external partner (matching in kind)



Department of Data Science and Knowledge Engineering

March 14th 2022

Subject: Letter of intent

Dear Dr Irene Grossman and Prof. Pieter van Gelder,
I, (Dr Rachel Cavill) hereby confirm that I intend to contribute to the *Patient-centered oncological care. New paradigms in understanding and treating cancer* project, as applied for by the main applicant Dr Irene Grossmann, Delft university of Technology.

I intend to provide an in-kind contribution of 0.1 FTE of my time.

Yours sincerely,

Dr Rachel Cavill
Assistant Professor

Visiting address
Paul-Henri Spaaklaan 1
6229 EN Maastricht

Postal address
P.O. Box 616
6200 MD Maastricht
The Netherlands

T +31 (0)43 388 3494
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BIC: INGBNL2A
VAT identifier EU
NL0034.75.268.B01

www.maastrichtuniversity.nl
<http://www.maastrichtuniversity.nl/web/Schools/DKE.htm>

KvK nr.: 50169181

Roche Diagnostics Nederland B.V.

DocuSign Envelope ID: B7DC0EAD-DD71-4204-BE31-0F3699FC7693



Erasmus MC
Pathology department
Dr. Jan van der Thüsen
Dr. Molewaterplein 40
3015 GD Rotterdam,
The Netherlands

Almere, March 14th 2022

Subject: Letter of intent Flagship program Convergence Health & Technology mission

Dear Dr Von der Thüsen,

With this letter, Roche Diagnostics Nederland B.V. expresses its intent to support the Flagship program Convergence Health & Technology mission proposal "**Patient-centered oncological care - New paradigms in understanding and treating cancer**" and the therein described ambitions to build an integrated methodology for personalised oncological care, combining predictive mathematical modelling with medical knowledge, real data, in-vitro tests, data science, machine learning, multi-objective decision making and healthcare management principles, to optimise its implementation in clinical practice.

I hereby confirm, Mrs. E.R. de Rooij, in my capacity as General Manager of Roche Diagnostics Nederland B.V., that Roche Diagnostics Nederland B.V., upon your specific request dated 09/02/2022 intends to contribute, in kind to the above mentioned proposal "**Patient-centered oncological care - New paradigms in understanding and treating cancer**" with 1-2 hours of our experts per week for the duration of the project and a potential cash contribution of € 40.000,-. These values are an estimation and will be further defined at a later stage depending on the development of the program(s).

Roche Diagnostics Nederland B.V. is very connected to the important "**Patient-centered oncological care**". We have extensive (national and international) experience in diagnostics and data which we are very open about and we are willing to share our knowledge and expertise on this topic. Once the scope and viability is addressed we could consider additional contributions in kind in the form of software or diagnostic tools.

This contribution is in the form of the above described expertise and sharing of knowledge and it is the intention to make this contribution available according to the further to be defined planning of the project proposal, under the condition that the investments described in the proposal actually take place and that the Flagship program makes the requested contribution.

Roche Diagnostics
Nederland BV

Postbus 1007
NL - 1300 BA ALMERE
Nederland

Handelsregister KvK
Revoland nr. 39035097
Roche Pharmaholding BV
IBAN: DE53120700700010006034
BIC/Swiftcode: DEUTDEFFVAC
btw nr. NL-00.14.15.037-B01

Tel. +31 (0)36 53 94 911

Mk

1/2

Roche Nederland B.V.



ErasmusMC,
J.H. von der Thüsen, MD PhD, pathologist,
Dr. Molewaterplein 40,
3015 CA Rotterdam

Woerden, March 8, 2022

Our ref.: 2022013MLSR

Subject: Patient-centered oncological care: new paradigms in understanding and treating cancer

Dear mr. Von der Thüsen,

Recently we discussed the concept of the project that you are developing to be submitted as a Flagship program for the program round 2022: "Patient-centered oncological care: new paradigms in understanding and treating cancer".

You have provided us with a draft application form of the project and the "Call for Proposals Flagship Programs Round 2022".

Based on our discussion and the information in the application form, with this letter, we inform you that Roche Nederland b.v. ('Roche') is interested in studying your research proposal further when this will be complete and available.

At this point no other commitment will be made and Roche is not legally nor financially bound by, or to, the submission and or the Project.

Yours sincerely,

Roche Nederland B.V.

Mrs. J. Timmermans - van Boxtel, Msc
Chapter Lead Customer Value Solutions

Roche Nederland b.v.
Beneluxlaan 2A 3446 GR
Woerden

Mrs. S. Rotteveel, Msc
Patient Journey Partner

III Letter of support from department head of the main lead

Date 11 March 2022
Our reference 22-113/SR/nvdh
Telephone + 31 15 27 88779
E-mail s.roeser@tudelft.nl
Subject Letter of support



Faculty Technology, Policy and Management
Department Values Technology and
Innovation

Visiting address
Jaffalaan 5 (building 31)
2628 BX Delft
Postal address
P.O. Box 5015
2600 GA Delft
The Netherlands

*LETTER OF SUPPORT
for the
Patient-centered oncological care.
New paradigms in understanding and treating cancer.
PROJECT*

I, Prof. Dr. Sabine Roeser, in my capacity of Head of the Department of Values, Technology, and Innovation at Delft University of Technology hereby confirm that we are aware of and support the flagship initiative *Patient-centered oncological care. New paradigms in understanding and treating cancer*, as applied for by the main applicant, Dr. Irene Grossmann, assistant professor at Delft University of Technology, TPM faculty.

Yours sincerely,

A handwritten signature in black ink, appearing to read "Sabine Roeser".

Name: Prof. Dr. Sabine Roeser

Position: Head of the Department of Values, Technology, and Innovation at TPM faculty,
Delft University of Technology

Date: March 11, 2022

IV References

- [1] Afghari, A. P., Haque, M. M., Washington, S., & Smyth, T. (2016). Bayesian latent class safety performance function for identifying motor vehicle crash black spots. *Transportation research record*, 2601(1), 90-98.
- [2] Anjum, R. L., Copeland, S., & Rocca, E. (2020). Rethinking causality, complexity and evidence for the unique patient: a causehealth resource for healthcare professionals and the clinical encounter- E-book. Springer Nature.
- [3] Archibald, M., Wiebe, S., Rieger, K., Linton, J., & Woodgate, R. (2021). Protocol for a systematic review of living labs in healthcare. *BMJ open*, 11(2), e039246. doi: 10.1136/bmjopen-2020-039246.
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